OIML Member State SLOVAKIA	**OIML Certificate No.** **R49/2024-A-SK1-26.03**

OIML CERTIFICATE ISSUED UNDER SCHEME A

OIML Issuing Authority

Name: **Slovak Legal Metrology (SLM)**
Address: Geologická 9966/1,
821 06 Bratislava-Podunajské Biskupice, Slovakia
Product Certification Body
Hviezdoslavova 31
974 01 Banská Bystrica, Slovakia
Person responsible: Ing. Dušan Šmigura, PhD., Director of PCB

Applicant

Name: **FineTek Co., Ltd.**
Address: No. 16, Ziqiang St., Tucheng Dist., New Taipei City,
23678, Taiwan, R.O.C.

Manufacturer

Name: **FineTek Co., Ltd.**
Address: No. 16, Ziqiang St., Tucheng Dist., New Taipei City,
23678, Taiwan, R.O.C.

Identification of the certified type *(the detailed characteristics are defined in the additional pages)*

Water meter type **EPD1 series**

Designation of the module *(if applicable)*

Electromagnetic water meters with electronic indication device

This OIML Certificate attests the conformity of the above identified type (represented by the sample(s) identified in the OIML type evaluation report) with the requirements of the following Recommendation of the International Organization of Legal Metrology (OIML):

OIML R 49, Edition (year): 2024
For accuracy class: 2

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R49/2024-A-SK1-26.03

This OIML Certificate relates only to metrological and technical characteristics of the type of measuring instrument covered by the relevant OIML Recommendation identified above.

This OIML Certificate does not bestow any form of legal international approval.

The conformity was established by the results of tests and examinations provided in the associated:

OIML type evaluation report No. 2026/ER064/SK1 dated 12th October 2026 that includes 28 pages.

The technical documentation relating to the identified type is contained in documentation file names:

- Technical documentation file FineTek_EPD1_00, dated 5th October 2026, includes 204 pages.

OIML Certificate History

Revision No.	Date	Description of the modification
0	12th October 2026	First certificate issuance (according to the OIML R49:2024).

Identification, signature and stamp

The OIML Issuing Authority

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Dušan Šmigura

Date: 12th October 2026

Important note: Apart from the mention of the Certificate's reference number and the name of the OIML Member State in which the Certificate is issued, partial quotation of the Certificate and of the associated OIML type evaluation report(s) is not permitted, although either may be reproduced in full.

1. Designation

The electromagnetic water meters **EPD1 series** (model EPD100 and EPD104) are designed to measuring, storing and displaying the volume of water passing through the measurement transducers at metering conditions. The water meter is intended for the measurement of volume of clean water in residential use.

Differences between models:
- model EPD100: indicating device is integral with the body;
- model EPD 104: indicating device is not integral with the body

The water meters EPD1 series are residential compact electromagnetic water meter with electronic indication device.
The measurement is based on electromagnetic principle.
According to Faraday's law of electromagnetic induction a conductive liquid flow through an insulated pipe, generating a voltage signal at electrodes on both sides of the pipe under the action of coil and is proportional to the fluid flow rate. The signal is amplified and filtered for accumulation - recording and output.
The water meters EPD1 series can be installed to operate in horizontal position. When the instrument is installed on the pipeline, the instrument must be equipped with a ground ring (Fig. 4).
The water meter is not designed to measure the reverse flow.

2. Description

2.1 Parts of the water meters EPD1 series

Essential parts of the water meters:
Flow sensor:
- the electromagnetic transducer - the steel body with inlet and outlet;
- two measurement electrodes;
- the coil for magnetic field generating a voltage signal at electrodes on both sides of the pipe;
- the grounding rings on both ends of the sensor must be installed in order for solid grounding.

Calculator and indication device:
- is mounted in a housing on flow sensor (EPD100);
- is mounted in in a housing separately from flow sensor (EPD104);
- the converter base connected with the electromagnetic transducer,
  The converter includes:
  - the PCBs
    - display board;
    - main board;
    - DC power board;
    - terminal board;
  - the graphical electronic LCD display integral with the body (EPD100) and not integral with the body-is integral in separately box (EPD104).
    The LCD display is with three lines:
    - **first line** with 5 digits and indication range 99999 m^3/h for flowrate,
    - **second line** with 5 or 6 or 7 digits and indication range 99999 or 999999 or 9999999 m^3 for volume, according to tables in point 3). The sub-multiples of a cubic meters are indicated on the LCD display after the decimal point and are marked with a line under numbers,
    - **third line** pressure in tube and L:.. H for ratio relative to the full-scale flow rate;
- DC voltage source;
- the power line, the 2-core copper conductor with rubber insulated, outer diameter within 8 mm to 12 mm and core wire size within 0.75~1.0 mm 2(20~16 AWG) cable;
- three operation buttons for display control.

Non-essential parts of the water meter:

- communication type RS-485;
- analogue output 4-20 mA, DO, DI, Pulse output.

2.2 Metrological functions

- measuring, memorizing and displaying the volume of water passing through the water meter.

2.3 Operation and presentation of legal data

The LCD display in user mode:

a) the total measured volume (m^3);
b) flow rate (m^3/h);
c) pressure in tube (kPa);
d) software version;
e) checksum (CRC version).

The following settings or presentation can be made using the buttons under admin password:

a) zero-flow compensation (m/s);
b) sensor characteristic factor k-Factor;
c) restore factory settings;
d) software version;
e) checksum (CRC version).

The LCD display includes following display screens when power on:

a) Test display – full screen;
b) Test display – blank screen;
c) Self-diagnosis of following components;
   - temperature sensor;
   - communication module;
   - measuring tube;
   - memory;
   - excitation coil.

2.4 Software specification

Software versions	Checksum	Remarks
v28	016D5D74	-

The software version is indicated on the LCD display in menu 8.2 in the form MV: v28.
The checksum is indicated on the LCD display in menu 8.4 in the form: Checksum=016D5D74.
Note: Firmware is in accordance with OIML D31: 2023/ OIML R49-1: 2024, section D.4.3.

2.5 Accountable alarms

Described in Operation manual.

3. Technical and metrological data

Type /model		EPD100 / EPD104										
Nominal diameter DN	mm	50	80	100	150	200	250	80	100	150	200	250
Permanent flowrate Q_3	m^3/h	63	160	250	630	1000	1600	40	63	160	250	400
Minimum flowrate Q_1	m^3/h	0,39375	1	1,5625	3,9375	6,25	10	1	1,575	4	6,25	10
Transitional flowrate Q_2	m^3/h	0,63	1,6	2,5	6,3	10	16	1,6	2,52	6,4	10	16

Overload flowrate Q_4	m^3/h	78,75	200	312,5	787,5	1250	2000	50	78,75	200	312,5	500
Ratio Q_3/Q_1	-	160						40				
Ratio Q_2/Q_1	-	1,6										
Connection thread	-	flange										
Construction length L	mm	200	200	250	300	350	400	200	250	300	350	400
Installation position	-	Horizontal										
Water temperature range	°C	0,1 to 30										
Meter temperature class	-	T30										
Maximum working pressure MAP	bar	10										
Pressure loss class ΔP	bar	0,10										
Maximum permissible error in upper flowrates range $Q_2 \leq Q \leq Q_4$	%	± 2 (at $\Theta \leq 30°C$) ± 3 (at $\Theta > 30°C$)										
Maximum permissible error in lower flowrates ranges $Q_1 \leq Q < Q_2$	%	± 5										
Scale interval	m^3	0.0001	0.0001	0.001	0.001	0.001	0.001	0.0001	0.001	0.001	0.001	0.001
Capacity of calculator	m^3	99999	999999	999999	999999	9999999	9999999	999999	999999	999999	9999999	9999999
Accuracy class	-	2										
Mechanical class	-	M1										
Climatic class	°C	-25 to +55										
Electromagnetic class	-	E1										
Environmental classification	-	O										
Flow profile sensitivity class	-	U0D0										

4. Marking and inscriptions

The following data shall be marked on the water meter:

a) unit of measurement (m^3);
b) numerical value of Q_3 and the ratio Q_3/Q_1:
c) type approval sign according to national regulations;
d) name or trademark of the manufacturer;
e) year of manufacture, the last two digits of the year of manufacture, or the month and year of manufacture;
f) serial number (as near as possible to the indicating device);
g) direction of flow, by means of an arrow (shown on both sides of the body or on one side only provided the direction of flow arrow is easily visible under all circumstances);
h) maximum admissible pressure (MAP) if it exceeds 1 MPa (10 bar);
i) letter H, if the meter can only be operated in the horizontal position;
j) the pressure loss class where it differs from Δp 63;
k) environmental classification;
l) electromagnetic environmental class.

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Designation of trademarks on the water meters

The manufacturer uses following trademarks on the water meter:

FineTek FineTek FineTek

5. Security measures

The water meter shall be protected against unauthorised manipulation and opening as follows:

- PCB Conformal Coating: The electronic components on the PCB are treated with conformal coating (three-proof coating), providing the electronic module with moisture resistance, dust protection, and antistatic performance.
- type EPD100 (compact / integral type): one wire-type lead seal is installed at the junction between the upper cover and the lower body of the water meter (which contains the flow sensor), preventing unauthorized access to the PCBA and firmware (Fig. 2a).
- type EPD104 (remote / separated type): one wire-type lead seal is installed at the junction between the upper and lower parts of the flow sensor. An additional wire-type lead seal, together with a dedicated copper post (sealing post), is installed on the separately-housed enclosure of the calculator and indication device, preventing unauthorized access to the PCBA and firmware (Fig. 2b).

6. Figures

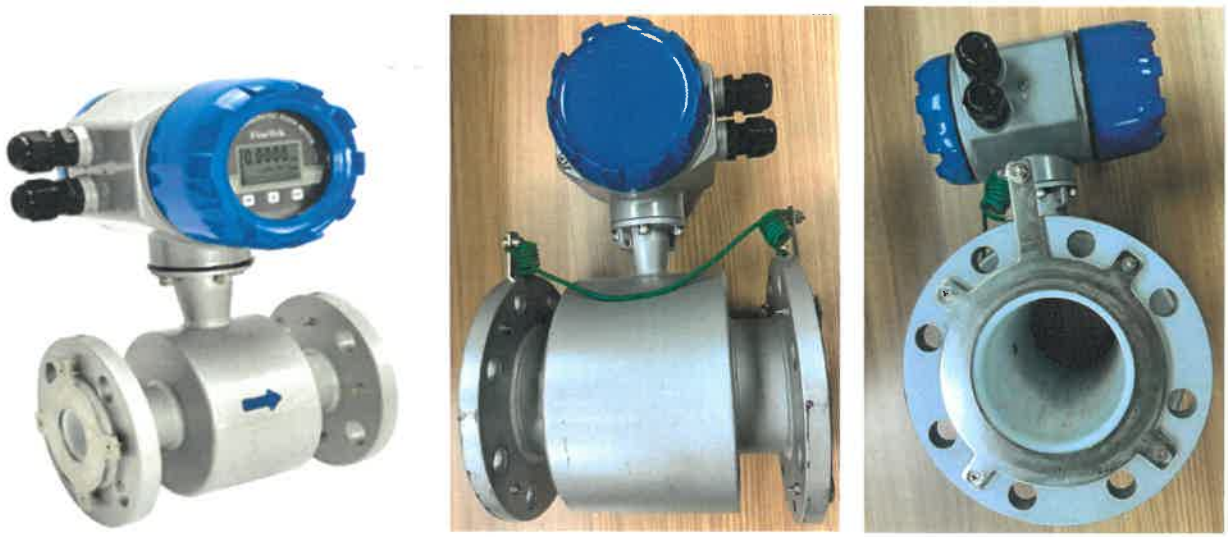


Fig. 1a: Illustrative views of the water meters type EPD100

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Fig. 1b: Illustrative views of the water meters type EPD104

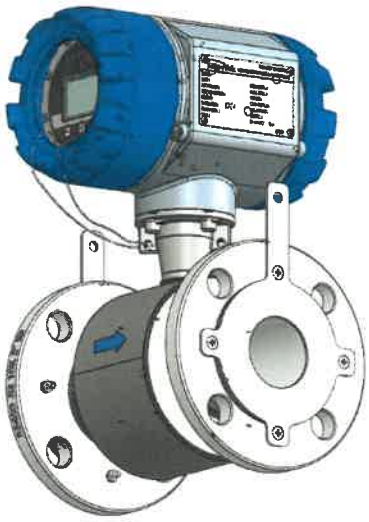


Fig. 2a: Sealing of the type EPD100

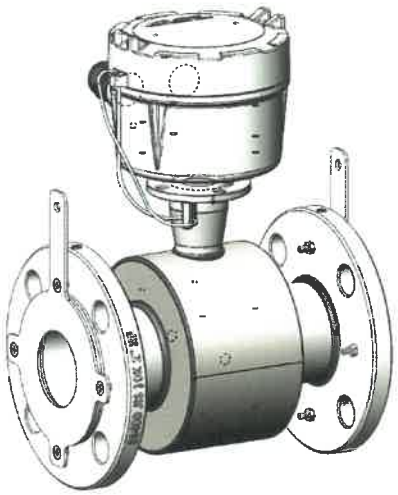




Fig. 2b: Sealing of the type EPD104

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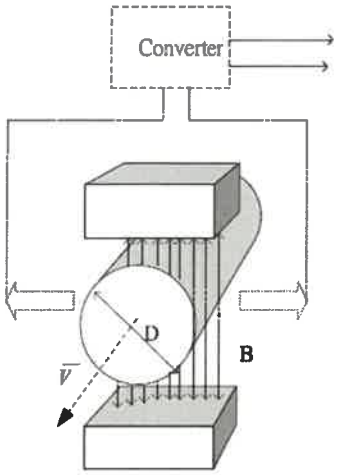


Fig. 3: The diagram of the internal structure

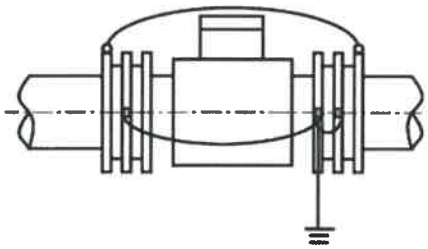


Fig. 4: Grounding diagram

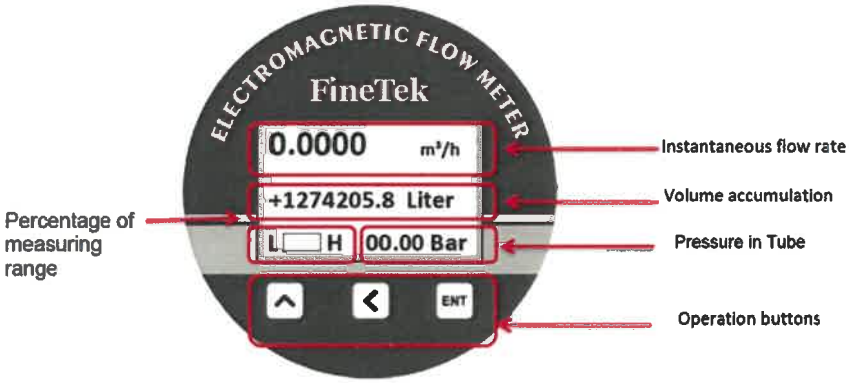


Fig. 5a: Display for EPD100

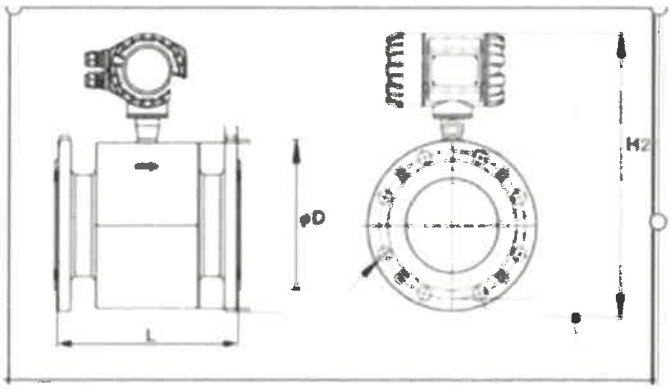


Fig. 5b: Real display with marked numbers after decimal point



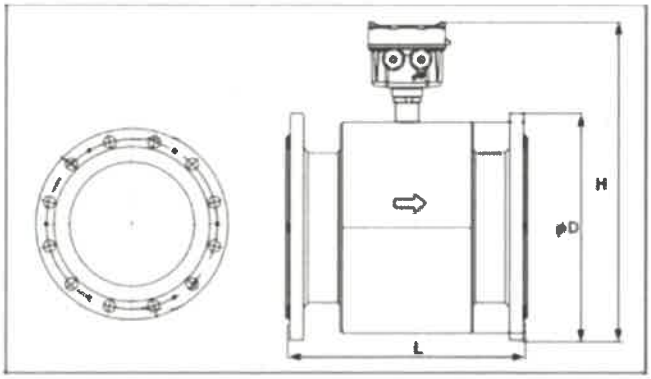
Fig. 5c: Display for EPD104

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DN	reference dimension			Weight (Kg)
	L (mm)	H (mm)	D (mm)	
50	200	322	155	7.7
80	200	347	185	9.9
100	250	385	210	12.9
150	300	467	280	23.5
200	350	506	330	33.2
250	400	572	400	54

Fig. 6a: Dimensions of the water meters type EPD100



DN	reference dimension			Weight (Kg)
	L (mm)	H (mm)	D (mm)	
50	200	284	155	6.1
80	200	309	185	8.3
100	250	346	210	11.4
150	300	429	280	22
200	350	468	330	31.7
250	400	534	400	52.5

Fig. 6b: Dimensions of the water meters type EPD104

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